Data Quality and Validation in ETL

Assignment Question

PW SKILLS

Data Quality and Validation in ETL

Question 1 : Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Question 2 : Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Question 3 : What is duplicate data? Explain three causes in ETL pipelines.

Question 4 : Differentiate between exact, partial, and fuzzy duplicates.

Question 5 : Why should data validation be performed during transformation rather than after loading?

Question 6 : Explain how business rules help in validating data accuracy. Give an example.

Use the following `Sales_Transactions` dataset for all questions:

Txn_ID	Customer_ID	Customer_Name	Product_ID	Quantity	Txn Amount	Txn_Date	City
201	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
202	C102	Anjali Rao	P12	1	1500	2025-12-01	Bengaluru
203	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai
204	C103	Suresh Iyer	P13	3	6000	2025-12-02	Chennai
205	C104	Neha Singh	P14	Null	2500	2025-12-02	Delhi
206	C105	N/A	P15	1	Null	2025-12-03	Pune
207	C106	Amit Verma	P16	1	1800	Null	Pune
208	C101	Rahul Mehta	P11	2	4000	2025-12-01	Mumbai

Question 7 : Write an SQL query on `Sales_Transactions` to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount)

Question 8 : Enforcing Referential Integrity
Assume the following `Customers_Master` table:

CustomerID	CustomerName	City
C101	Rahul Mehta	Mumbai
C102	Anjali Rao	Bengaluru
C103	Suresh Iyer	Chennai
C104	Neha Singh	Delhi

Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations.